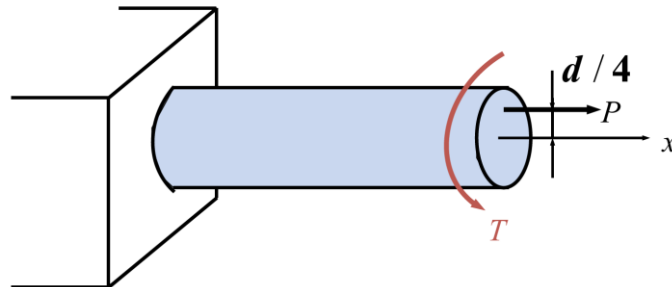


Submission: Please submit the scanned/typed document to Canvas.

1. A solid circular shaft of diameter d is subjected to a torsional moment T and an axial force P . Given that $T = Pd$, identify the location of the critical point (the point with highest principal stress σ_1) and write the corresponding principal stress σ_1



Solution:

Replace the eccentric load with an equivalent tensile load and a bending moment.



Tensile force P

$$\sigma_{x1} = \frac{P}{A} = \frac{4P}{\pi d^2}$$

Bending moment M

$$\sigma_{x2} = -\frac{My}{I} = -\frac{\left(-\frac{Pd}{4}\right)y}{\frac{\pi d^4}{64}} = \frac{16Py}{\pi d^3}$$

Torsional moment T

$$\tau = \frac{T\rho}{I_p} = \frac{(Pd)\rho}{\frac{\pi d^4}{32}} = \frac{32P\rho}{\pi d^3}$$

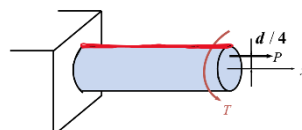
The maximum principal stress:

$$\sigma_1 = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau^2} = \frac{1}{2}\left(\frac{4P}{\pi d^2} + \frac{16Py}{\pi d^3}\right) + \sqrt{\frac{1}{4}\left(\frac{4P}{\pi d^2} + \frac{16Py}{\pi d^3}\right)^2 + \left(\frac{32P\rho}{\pi d^3}\right)^2}$$

When $y = \frac{d}{2}, \rho = \frac{d}{2}$, σ_1 reaches its maximum value

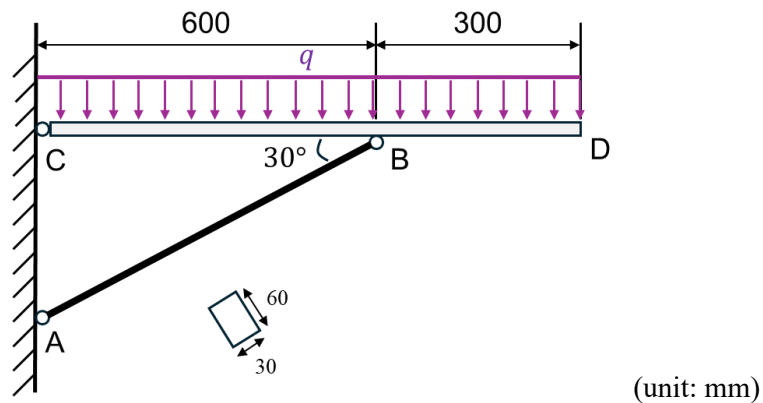
$$\sigma_1 = (2\sqrt{73} + 6) \frac{P}{\pi d^2}$$

The critical point is



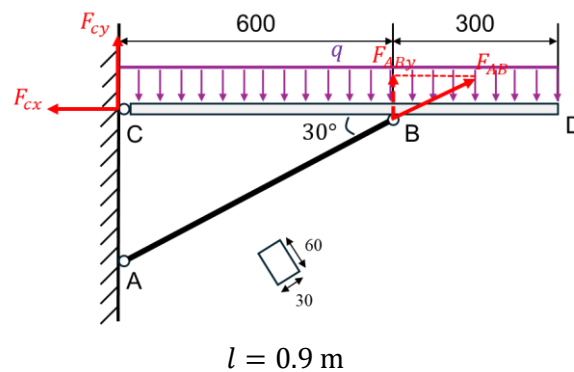
2. A solid circular beam CD has a diameter of $d = 100$ mm, $[\sigma] = 160$ MPa. Bar AB with a rectangular cross section, $E = 200$ GPa, $\sigma_{pl} = 200$ MPa, $\sigma_s = 240$ MPa. The linear relation of inelastic buckling has coefficient $a = 310$ MPa and $b = 1.14$ MPa, The safety factor of stability is $n_{st} = 3$. Ignore the transverse shear and axial force in beam CD. Please determine the allowable load q_{max} .

(Hint: the force acting on the bar AB is along its longitudinal axis)



Solution:

FBD:

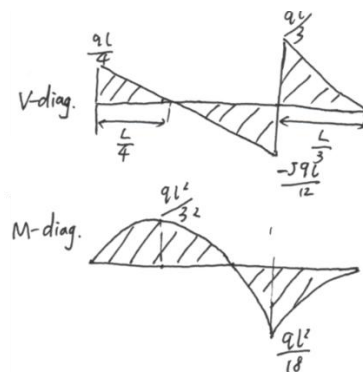


Equilibrium equation

$$M_C = 0 \Rightarrow ql \cdot \frac{l}{2} = F_{AB} \cdot \frac{2}{3}l \sin 30^\circ \Rightarrow F_{AB} = \frac{3}{2}ql$$

$$F_{cy} + F_{AB} \sin 30^\circ = ql \Rightarrow F_{cy} = \frac{ql}{4}$$

V-diag and M-diag using Jump method



The maximum moment in beam CD

$$|M_{max-CD}| = \frac{ql^2}{18}$$

By the strength of CD

$$|\sigma_{max-CD}| = \frac{|M_{max-CD}| \left(\frac{1}{2}d\right)}{\left(\frac{\pi d^4}{64}\right)} \leq [\sigma] \Rightarrow q \leq 349.1 \text{ kN/m}$$

Check the stability of AB (pinned-pinned, $K = 1$)

The column will bulking along the direction with smaller I , the slenderness ratio

$$\lambda = \frac{KL}{\sqrt{I_{small}/A}} = \frac{L}{\sqrt{\frac{b^2}{12}}} = \frac{\frac{2}{3}l}{b\sqrt{\frac{1}{12}}} = 80$$

$$\lambda_{pl} = \sqrt{\frac{\pi^2 E}{\sigma_{pl}}} = 99.35$$

$$\lambda_s = \frac{a - \sigma_s}{b} = 61.4$$

So $\lambda_s < \lambda < \lambda_p$, intermediate column

$$\sigma_{cr} = a - b\lambda = 218.8 \text{ MPa}$$

The stability condition:

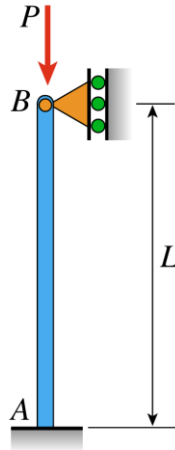
$$\frac{F_{AB}}{A} \leq \frac{\sigma_{cr}}{n_{st}}$$

$$\Rightarrow \frac{3}{2}ql \leq \frac{\sigma_{cr}A}{3} = 131.28 \text{ kN} \Rightarrow q \leq 97.2 \text{ kN/m}$$

So the allowable load

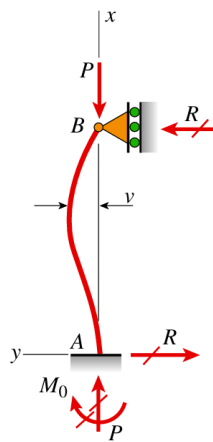
$$q_{\max} = 97.2 \text{ kN/m}$$

3. Please determine the critical buckling load of an initially straight prismatic column AB with length L , Young's modulus E , in-plane moment of inertia I . The bottom end, B , is fixed to the ground, while the top end, A , is pinned to a roller that is free to move vertically.



Solutions:

We determine the critical buckling load, P_{cr} , by finding the loads at which infinite nontrivial neutral equilibrium solutions exist.



From the equilibrium of the entire column, there must be horizontal reactions R at each end such that

$$M_0 = RL$$

The bending moment in the buckled column, at distance x from the base, is

$$M = M_0 - Pv - Rx = -Pv + R(L - x)$$

Therefore, the differential equation is

$$EIv'' = M = -Pv + R(L - x)$$

Substitute $k^2 = P/EI$ and rearrange to get

$$v'' + k^2 v = \frac{R}{EI}(L - x) \quad (1)$$

(1) The homogeneous equation:

$$v'' + k^2 v = 0$$

Seek $v = e^{rx}$ for the solution

$$r^2 e^{rx} + k^2 e^{rx} = 0 \Rightarrow r = \pm ik$$

The general solution

$$\begin{aligned} v(x) &= A_1 e^{ikx} + A_2 e^{-ikx} = (a_1 + ib_1)(\cos kx + i \sin kx) + (a_2 + ib_2)(\cos kx - i \sin kx) \\ &= (a_1 + a_2) \cos kx + (b_2 - b_1) \sin kx + i(\dots) \end{aligned}$$

Take the real part and rename the coefficients

$$v(x) = C_1 \sin kx + C_2 \cos kx$$

(2) The particular solution:

$$k^2 v = \frac{R}{EI}(L - x) \Rightarrow v(x) = \frac{R}{P}(L - x)$$

The general solution for equation (1) is

$$v(x) = C_1 \sin kx + C_2 \cos kx + \frac{R}{P}(L - x)$$

The boundary conditions

$$v(0) = 0, \quad v(L) = 0, \quad v'(0) = 0$$

gives

$$C_2 + \frac{RL}{P} = 0, \quad C_1 \sin kL + C_2 \cos kL = 0, \quad C_1 k - \frac{R}{P} = 0 \quad (2)$$

To get non-zero solution, eliminate R from the first and third equations

$$\Rightarrow C_2 = -C_1 kL \quad (3)$$

Substitute (3) into the second equation in (2)

$$kL = \tan kL$$

The smallest nonzero value of kL is 4.4934.

The corresponding critical load is

$$\begin{aligned} \frac{P}{EI} &= \left(\frac{4.4934}{L} \right)^2 \\ P_{cr} &= \frac{20.19EI}{L^2} = \frac{\pi^2 EI}{(0.669L)^2} \end{aligned}$$